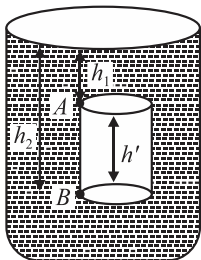
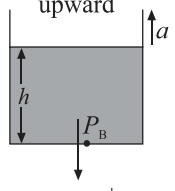
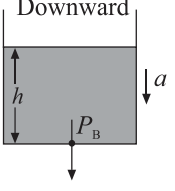
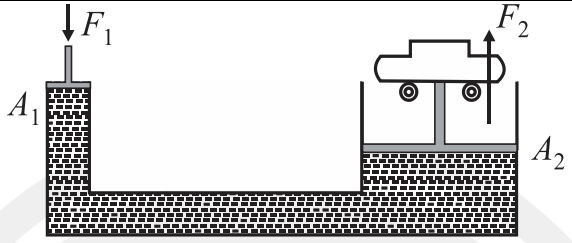
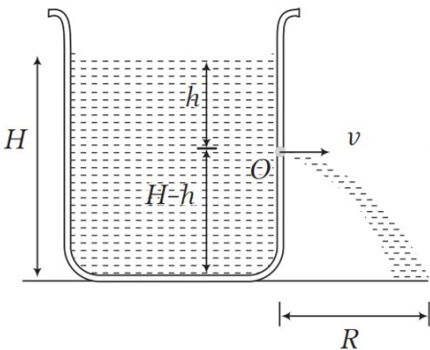


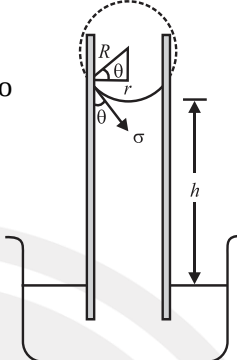
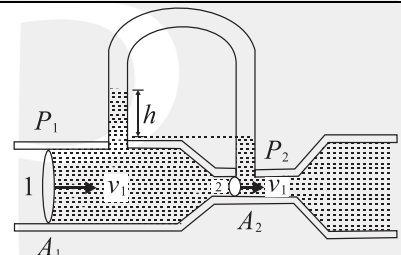
Mechanical Properties of Fluids

Sr. No.	Concept	Formulas	Other information
1.	Thrust and pressure	$\text{Pressure} = \frac{\text{Thrust}}{\text{Area}} \text{ or } P = \frac{F}{A}$ $\text{If area is very small, then } P = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A} = \frac{dF}{dA}$	A = surface area. F = normal force acting on the surface.
2.	Density and relative density	$\text{Density, } \rho = \frac{\text{Mass (M)}}{\text{Volume (V)}} = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV}$ $\text{Relative density} = \frac{\text{Density of liquid or substance}}{\text{Density of water at } 4^\circ\text{C}}$	M = mass of the body. V = volume of the body.
3.	Density of a mixture of two or more liquids	If liquids of different volumes are mixed: $\rho = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{V_1 + V_2} = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2}$ If liquids of different masses are mixed: $\rho = \frac{(m_1 + m_2)}{(V_1 + V_2)} = \frac{(m_1 + m_2)}{\left(\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}\right)}$	ρ = density of the mixture. ρ_1, ρ_2 = densities of two liquids. m_1, m_2 = masses of two liquids. V_1, V_2 = volumes of two liquids.
4.	Variation of fluid pressure with depth	Total pressure at a depth h in liquid: $P_{\text{total}} = P_0 (\text{atmospheric pressure}) + \rho gh$  $\therefore$ Pressure difference $P_B - P_A = (P_0 + h_2 \rho g) - (P_0 + h_1 \rho g) = (h_2 - h_1) \rho g$ $P_B - P_A = h' \rho g$	P_A = pressure at point A. P_B = pressure at point B. ρ = density of liquid. h = height of the liquid column. P_0 = pressure exerted by 76 cm of mercury column = $1.013 \times 10^5 \text{ N/m}^2$

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5.	Variation of pressure in an accelerated container	Container moving upward  $g_{\text{eff}} = g + a$ $P_B = P_0 + \rho(g + a)h$ Container moving Downward  $g_{\text{eff}} = g - a$ $P_B = P_0 + \rho(g - a)h$	P_0 = atmospheric pressure g = gravitational acceleration a = acceleration of the container ρ = density of liquid in container
6.	Pascal's law and Hydraulic lift	 $P = \frac{F_1}{A_1} = \frac{F_2}{A_2} \text{ or } F_2 = \frac{A_2}{A_1} \cdot F_1$ $A_2 \gg A_1, \text{ therefore } F_2 \gg F_1.$	A_1, A_2 = area of cross-sections of the two ends of the U-shaped tube. F_1, F_2 = effective force on area of the two ends of the tube. P = pressure.
7.	Archimedes' principle	Apparent weight of a body in a fluid = True weight – Weight of fluid displaced $W_{\text{app}} = W - U = V\sigma g - V\rho g$ $= V\sigma g \left(1 - \frac{\rho}{\sigma}\right) = mg \left(1 - \frac{\rho}{\sigma}\right)$	V = volume of the body or displaced fluid. m = mass of the body ρ = density of liquid σ = density of body.
8.	Buoyant force (upward thrust)	Buoyant force or upthrust force on the body; F = weight of liquid displaced = $V\rho g$	V = volume of the displaced liquid. ρ = density of liquid.
9.	Law of floatation	When a body is floating; Weight of the body = Weight of liquid displaced $V\sigma g = V'\rho g \text{ or } \frac{V'}{V} = \frac{\sigma}{\rho}$	V' = volume of displaced fluid. V = volume of the body. g = gravitational acceleration.
10.	Coefficient of viscosity	Velocity gradient between two layers of liquid = $\frac{dv}{dx}$ Newton's formula for viscous force between two parallel layers is; $F = -\eta A \frac{dv}{dx}$ $\Rightarrow \eta = \frac{-F}{A \left(\frac{dv}{dx}\right)}$	η = coefficient of viscosity. F = viscous force. A = area of the layers.

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11.	Poiseuille's formula	The volume of a liquid flowing out per second: $Q = \frac{V}{t} = \frac{\pi P r^4}{8 \eta l}$	r = radius of the capillary tube. l = length of the capillary tube. P = pressure difference across the two ends of the tube. V = volume of the liquid flown. t = time.
12.	Stoke's law	The backward dragging force acting on body: $F = 6\pi\eta r v$	η = coefficient of viscosity of the medium. r = radius of the spherical body. v = velocity of the body. F = viscous force.
13.	Terminal velocity (V_T)	Maximum constant velocity attained by a spherical body: $V_T = \frac{2r^2}{9\eta}(\rho - \sigma)g$	ρ = density of the falling body. σ = density of fluid. r = radius of the body.
14.	Equation of continuity	For an incompressible and non-viscous liquid: $A_1 v_1 = A_2 v_2$ $Av = \text{constant}$ or $v \propto \frac{1}{A}$	A_1, A_2 = area of cross-sections of the two ends of the tube. v_1, v_2 = velocity of liquid at the two ends of the tube.
15.	Bernoulli's theorem	Bernoulli's theorem, $\frac{P}{\rho} + gh + \frac{1}{2}v^2 = \text{constant}$ One another form, $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$ Another form of Bernoulli's theorem, $\frac{P}{\rho g} + h + \frac{v^2}{2g} = \text{constant}$ or, Pressure head + Gravitational head + Velocity head = constant.	P = pressure at test point. ρ = density of liquid $\frac{P}{\rho g}$ = pressure head h = gravitational head $\frac{v^2}{2g}$ = velocity head

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16.	Efflux velocity (Torricelli's theorem)	The speed of efflux is: $v = \sqrt{2gh}$  The range of liquid: $R = 2\sqrt{h(H-h)}$ R is maximum at $h = \frac{H}{2}$ $R_{\max} = 2\sqrt{\frac{H}{2}\left(H - \frac{H}{2}\right)} \Rightarrow \boxed{R_{\max} = H}$ Time taken to empty a tank: $t = \frac{A}{a}\sqrt{\frac{2H}{g}}$	v = speed of the liquid. R = range of falling liquid. H = height of the liquid level. h = height of free surface of liquid above orifice. A = area of cross-section of the tank. a = area of hole.
17.	Surface tension	Surface tension = $\frac{\text{Force}}{\text{Length}}$ or $S = \frac{F}{l}$	l = length of the imaginary line. F = total cohesive force.
18.	Surface energy	Surface energy = $S \times \text{increase in area}$	S = surface tension
19.	Excess pressure in drops & bubbles	Inside a liquid drop, $P = \frac{2S}{R}$ (with one free surface) Inside a soap bubble, $P = \frac{4S}{R}$ (with two free surfaces) In an air bubble, $P = \frac{2S}{R}$ (with one free surface)	P = excess pressure. R = radius of the bubble. S = surface tension

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20.	Capillarity (Rise of liquid in a capillary tube)	The liquid rises or falls through a distance: $h = \frac{2S \cos \theta}{r \rho g}$ Rise of liquid in a tube of insufficient height: $h = \frac{2S}{R \rho g}$ As, σ, ρ and g are constants, so $hR = \frac{2\sigma}{\rho g} = \text{constant}$ 	θ = the angle of contact. r = radius of tube. ρ = density of liquid. S = surface tension. g = gravitational acceleration.
21.	Venturimeter	 We apply Bernoulli's equation; $P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$ From the continuity equation, $v_2 = \frac{A_1 v_1}{A_2}$ Substituting and rearranging, we get, $P_1 - P_2 = \frac{1}{2} \rho v_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right) = h \rho g$ Velocity of liquid flowing out per second; $v_1 = \frac{\sqrt{2gh}}{\sqrt{\left(\frac{A_1}{A_2}\right)^2 - 1}}$ Rate of flow, $A_1 v_1 = A_2 v_2 = A_1 \frac{\sqrt{2gh}}{\sqrt{\left(\frac{A_1}{A_2}\right)^2 - 1}}$	A_1, A_2 = area of cross-sections of the wider and narrow part of tube v_1, v_2 = velocity of liquid at the wider and narrow part of tube. P_1, P_2 = pressure at the wider and narrow part of tube. ρ = density of the liquid.